PRACTICAL LAB REPORT

Network Security and Security Operations Center Laboratory

WAZUH SIEM & XDR SECURITY OPERATIONS

Server Architecture, Multi-OS Agent Deployment, File Integrity Monitoring (FIM), Vulnerability Detection, Rootcheck, MITRE ATT&CK Mapping, and Automated Active Response

Practical Lab Report — XI

Submitted by	Mayank Raj
Programme	Master of Computer Applications (MCA)
Subject	Network Security Laboratory (Endpoint Detection & SIEM/XDR)
Experiment No.	11 — Wazuh Open-Source SIEM & XDR Home Lab Infrastructure
Tools & Environment	Wazuh 4.8.x All-in-One Server, Wazuh Agents (Linux & Windows), Kali Linux (Host: 10.204.133.156, MAC: 28:0c:50:ec:10:0f)
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LABORATORY DECLARATION

I hereby declare that the endpoint detections, file tampering verifications, vulnerability scans, rootkit checks, MITRE ATT&CK alignments, and automated active response scripts detailed in this report were executed entirely within a self-hosted, virtualized laboratory cluster. All endpoint modifications, simulated rootkit injections, and brute-force simulations were executed systematically by me for educational and analytical purposes. No external systems or uncontrolled enterprise environments were monitored or altered without prior authorization.

Laboratory Progression Series: Nmap Reconnaissance | Nessus Audit | Snort IDS | Wireshark DPI | Splunk SIEM | Wazuh XDR Lab

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1. Introduction & Security Engineering Objectives

Traditional SIEM tools excel at ingesting and aggregating normalized event logs from existing software daemons. However, detecting in-memory exploitation, file system manipulation, dormant kernel rootkits, and unpatched package vulnerabilities requires dedicated endpoint-level instrumentation. Wazuh provides an open-source Extended Detection and Response (XDR) and SIEM architecture capable of unifying endpoint-level telemetry with host intrusion detection (HIDS), compliance auditing, and autonomous remediation.

1.1 Specific Operational Objectives

- **Server Deployment:** Stand up an all-in-one Wazuh Central Node comprising the Wazuh Indexer, Wazuh Manager engine, and Wazuh Dashboard.
- **Cross-Platform Agent Enrollment:** Securely register and enroll both Linux (Ubuntu) and Windows Server endpoints via

1.2 Lab Infrastructure & Multi-OS Topology

The home lab architecture comprises three distinct virtual instances connected via an isolated local subnet (`10.204.133.0/24`). The analysis host executes the centralized Wazuh processing cluster, while physical and virtual agents simulate target enterprise infrastructure.

Host Identity	OS & Platform	IP Address	Functional Role & Software Components
Wazuh Central Host	Kali Linux 2026.x (x86_64)	`10.204.133.156`	Wazuh Manager 4.8, Wazuh Indexer (OpenSearch), Wazuh Dashboard
Linux Agent (Ubuntu)	Ubuntu Server 22.04 LTS	`10.204.133.180`	Wazuh Agent 4.8, OpenSSH, Apache2, Auditd, Syscheck, Rootcheck
Windows Agent	Windows Server 2022	`10.204.133.195`	Wazuh Windows Agent 4.8, Sysmon, Active Directory Domain Services
Adversary Node	Kali Linux (Attack Node)	`10.204.133.205`	Hydra v9.5, Metasploit Framework, Custom File Tampering Harness

2. Wazuh Central Architecture & Server Deployment

The Wazuh architecture splits operations across three decoupled planes: the *Wazuh Indexer* (a highly scalable OpenSearch-based document search engine), the *Wazuh Manager* (the core analytical daemon responsible for decoding logs, executing rulesets, and maintaining agent cryptographic keys), and the *Wazuh Dashboard* (the web-based graphical visualizer).

LISTING 2.1: INSTALLING AND INITIALIZING WAZUH SERVER STACK

```
     # Execute unattended all-in-one assistant installation script kali@kali:~$ curl -sO https://p
ackages.wazuh.com/4.8/wazuh-install.sh kali@kali:~$ sudo bash ./wazuh-install.sh -a -i [INFO]: ---
Installation Summary --- [INFO]: Wazuh indexer: Installed and started. [INFO]: Wazuh manager: Insta
lled and started. [INFO]: Wazuh dashboard: Installed and started. [INFO]: Web UI: https://10.204.13
3.156 [INFO]: User: admin | Password: [AUTOGENERATED_SECURE_TOKEN] [INFO]: Installation finished su
ccessfully. # Verify systemd service status kali@kali:~$ sudo systemctl is-active wazuh-manager waz
uh-indexer wazuh-dashboard active active active
```

2.2 Manager Port Architecture & Socket Verification

The manager listens on specific network ports to handle agent registration, cryptographic telemetry streams, and REST API calls. Socket bindings were audited using `ss -tulpn` to ensure secure operational readiness.

Port / Protocol	Process Name	Functional Description
`1514 / TCP & UDP`	`wazuh-remoted`	Receives encrypted event telemetry, FIM records, and alerts from enrolled agents.
`1515 / TCP`	`wazuh-authd`	Agent enrollment service for automated key distribution and registration.
`55000 / TCP`	`wazuh-apid`	HTTPS RESTful API enabling CLI tools and Dashboard control of the Manager.
`9200 / TCP`	`wazuh-indexer`	Encrypted REST interface for OpenSearch document indexing and retrieval.
`443 / TCP`	`wazuh-dashboard`	HTTPS web user interface accessed by SOC analysts.

3. Multi-OS Endpoint Agent Enrollment

Wazuh utilizes an endpoint-centric model. Each protected host executes an unprivileged agent that gathers system events, monitors file state, captures audit logs, and receives active response directives from the Manager.

3.1 Linux Agent (Debian/Ubuntu) Registration

The Linux agent was enrolled using the Manager's enrollment daemon (`wazuh-authd`). The registration command automatically retrieves a unique pre-shared AES key, binding the agent's ID to the central server.

LISTING 3.1: ENROLLING UBUNTU LINUX AGENT VIA CLI

```
     ubuntu@target-srv:~$ wget -qO - https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dea
rmor -o /usr/share/keyrings/wazuh.gpg ubuntu@target-srv:~$ echo "deb [signed-by=/usr/share/keyring
s/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wa
zuh.list ubuntu@target-srv:~$ sudo apt update && sudo WAZUH_MANAGER="10.204.133.156" WAZUH_AGENT_NA
ME="ubuntu-srv-01" apt install -y wazuh-agent ubuntu@target-srv:~$ sudo systemctl daemon-reload ubu
ntu@target-srv:~$ sudo systemctl enable --now wazuh-agent ubuntu@target-srv:~$ sudo /var/ossec/bin/
agent-auth -m 10.204.133.156 -A ubuntu-srv-01 2026/09/20 12:04:18 agent-auth: INFO: Registering age
nt to 10.204.133.156:1515 2026/09/20 12:04:18 agent-auth: INFO: Valid key received. Registered with
ID: 001
```

> **Key Management:** The cryptographic key generated for agent `001` is stored at `/var/ossec/etc/client.keys`, containing the agent ID, hostname, IP address, and 128-bit symmetric encryption key used for all ongoing TLS-encapsulated telemetry.

3.2 Windows Server Agent Deployment via PowerShell

On Windows Server 2022, the Wazuh agent is packaged as an MSI installer. Enrollment was executed via an administrative PowerShell session specifying the Manager IP address and descriptive agent identifier.

LISTING 3.2: AUTOMATED POWERSHELL ENROLLMENT ON WINDOWS SERVER

```
PS C:\Users\Administrator> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.8.0-1.msi -OutFile ${env:TEMP}\wazuh-agent.msi PS C:\Users\Administrator> Start-Process msiexec.exe -Wait -ArgumentList '/qn /i ${env:TEMP}\wazuh-agent.msi WAZUH_MANAGER="10.204.133.156" WAZUH_AGENT_NAME="win-ad-srv01"' PS C:\Users\Administrator> Start-Service -Name "Wazuh" PS C:\Users\Administrator> Get-Service -Name "Wazuh" Status Name DisplayName ------ ---- ----------- Running Wazuh Wazuh Windows Agent
```

3.3 Verification of Active Agent Cluster

To verify cross-platform enrollment from the command line, the Manager utility `manage_agents -l` was queried, listing both registered endpoints along with their assigned internal identifiers and dynamic session statuses.

LISTING 3.3: MANAGER CLI AGENT INVENTORY VALIDATION

```
kali@kali:~$ sudo /var/ossec/bin/manage_agents -l Available agents: ID: 001, Name: ubuntu-srv-01, IP: 10.204.133.180, Active ID: 002, Name: win-ad-srv01, IP: 10.204.133.195, Active
```

Agent ID	Endpoint Name	Platform & Kernel	Telemetry Modules Enabled
001	ubuntu-srv-01	Linux 5.15.0-105-generic (Ubuntu 22.04)	Syscheck (FIM), Rootcheck, SCA, Logcollector, Active-Response
002	win-ad-srv01	Windows Server 2022 Datacenter (10.0.20348)	Syscheck (FIM), Windows Eventlog, Sysmon, SCA, Active-Response

Both agents reported an `Active` operational heartbeat, completing phase one of the endpoint deployment.

4. File Integrity Monitoring (FIM) & Real-Time Tamper Detection

The Wazuh File Integrity Monitoring (FIM) daemon, known internally as `syscheck`, continuously audits file system changes. It generates cryptographic hash baselines (MD5, SHA1, and SHA256) of watched paths, monitoring for unauthorized additions, modifications, permissions drift, and deletions.

4.1 Syscheck Configuration (ossec.conf)

On the Linux agent, `syscheck` was configured within `/var/ossec/etc/ossec.conf` to monitor high-security configuration directories using the `realtime` attribute and Linux `auditd`-backed `whodata` tracking.

LISTING 4.1: SYSCHECK XML CONFIGURATION BLOCK

```
<syscheck> <disabled>no</disabled> <frequency>300</frequency>
<scan_on_start>yes</scan_on_start> <!-- Monitor critical system config directories in real time
with whodata --> <directories realtime="yes" check_all="yes"
whodata="yes">/etc/ssh,/etc/pam.d,/etc/shadow</directories> <directories realtime="yes"
check_all="yes">/var/www/html</directories> <ignore>/etc/mtab</ignore> </syscheck>
```

4.2 Validating File Tampering with Adversary Simulation

To test the alerting pipeline, a malicious SSH configuration modification was simulated by appending an unauthorized root-access directive into `/etc/ssh/sshd_config`.

LISTING 4.2: TERMINAL TAMPERING TEST ON UBUNTU AGENT

```
ubuntu@target-srv:~$ echo "PermitRootLogin yes" | sudo tee -a /etc/ssh/sshd_config ubuntu@tar
get-srv:~$ sudo touch /etc/ssh/backdoor_key.pub
```

Alert ID / Rule	Level / Severity	Targeted File Path	Observed Behavioral Anomaly
`Rule 550`	LEVEL 7 (HIGH)	`/etc/ssh/sshd_config`	Integrity checksum changed: SHA256 modified; Size increased (+20 bytes).
`Rule 554`	LEVEL 5 (MEDIUM)	`/etc/ssh/backdoor_key.pub`	New file added to critical system directory `/etc/ssh`.
`Rule 553`	LEVEL 7 (HIGH)	`/etc/shadow`	Whodata extraction identified `uid=0 (root)` executing `usermod`.

5. Automated Vulnerability Detection & CVE Management

Wazuh incorporates a dedicated *Vulnerability Detector* module within the Manager engine. Rather than executing active, intrusive network port probes (such as Nessus or OpenVAS), the module correlates software package manifests collected by endpoint agents against continuously updated CVE repositories from Canonical, Debian, Microsoft, and the NIST National Vulnerability Database (NVD).

5.1 Enabling Vulnerability Detector on the Manager

The vulnerability engine was configured within `/var/ossec/etc/ossec.conf` on the Manager, pulling security feeds for Ubuntu Jammy and Windows platforms:

LISTING 5.1: ENABLING VULNERABILITY FEEDS IN MANAGER CONFIGURATION

```
<vulnerability-detector> <enabled>yes</enabled> <interval>5m</interval>
<min_full_scan_interval>6h</min_full_scan_interval> <run_on_start>yes</run_on_start> <provider
name="canonical"> <enabled>yes</enabled> <os>jammy</os> <update_interval>1h</update_interval>
</provider> <provider name="nvd"> <enabled>yes</enabled> <update_interval>1h</update_interval>
</provider> </vulnerability-detector>
```

5.2 Audit Results & Patch Deficit Identification

Within minutes of initial synchronization, the Vulnerability Detector processed agent `001`'s software inventory (1,240 packages) and generated immediate patch alerts for unpatched dependencies:

CVE Identifier	Package Name	Installed Version	CVSS v3	Severity	Remediation Status
CVE-2023-38408	openssh-server	1:8.9p1-3ubuntu0.1	9.8	CRITICAL	Upgrade to `openssh-server >= 1:8.9p1-3ubuntu0.4`
CVE-2023-44487	apache2	2.4.52-1ubuntu4.2	7.5	HIGH	HTTP/2 Rapid Reset Mitigation: Apply package update
CVE-2023-32233	linux-image-generic	5.15.0-70-generic	7.8	HIGH	Netfilter nf_tables privilege escalation: Kernel patch required
CVE-2023-4911	libc6	2.35-0ubuntu3.1	7.8	HIGH	Looney Tunables Glibc buffer overflow: Upgrade glibc

6. Rootkit Detection (Rootcheck) & Baseline Hardening

The *Rootcheck* engine executes specialized host anomaly scans to detect covert rootkits, hidden files, anomalous promiscuous interfaces, and undocumented listening ports. It runs concurrently with Security Configuration Assessment (SCA) to gauge compliance against published CIS (Center for Internet Security) benchmarks.

6.1 Rootcheck Engine Configuration

The module uses signature definition files (such as `rootkit_trojans.txt` and `rootkit_files.txt`) to search for indicators of known Linux kernel rootkits (e.g., Diamorphine, adore-ng).

LISTING 6.1: ROOTCHECK ENGINE DIRECTIVES IN OSSEC.CONF

```
<rootcheck> <disabled>no</disabled> <check_files>yes</check_files>
<check_trojans>yes</check_trojans> <check_dev>yes</check_dev> <check_sys>yes</check_sys>
<check_pids>yes</check_pids> <check_ports>yes</check_ports>
<check_interfaces>yes</check_interfaces> </rootcheck>
```

6.2 Executing Rootkit Sweeps & CIS Benchmarking

A test environment check was performed by creating an anomalous hidden file in `/dev` and simulating a concealed process:

LISTING 6.2: SIMULATING MALICIOUS ROOTKIT INDICATORS

```
ubuntu@target-srv:~$ sudo touch /dev/.hidden_backdoor ubuntu@target-srv:~$ sudo /var/ossec/bi
n/rootcheck -t
```

Module Engine	Alert Description	Detection Mechanism / Path	Remediation Guidance
Rootcheck	Hidden file found in `/dev`	Found `/dev/.hidden_backdoor`. Standalone files in `/dev` are suspicious.	Investigate process owning inode; isolate and remove file.
Rootcheck	Promiscuous mode enabled	Interface `eth0` operating in promiscuous listening mode.	Verify whether legitimate packet sniffer (Wireshark) is active.
CIS / SCA	SSH root login permitted	Policy Rule: Verify `PermitRootLogin` is set to no.	Update `/etc/ssh/sshd_config` and reload service.
CIS / SCA	Core dumps unrestricted	Policy Rule: Ensure core dumps are restricted via `limits.conf`.	Set `* hard core 0` in `/etc/security/limits.conf`.

7. MITRE ATT&CK Framework Tactical Mapping

A key architectural capability of Wazuh is the automated contextual mapping of decoder alerts to the MITRE ATT&CK framework. Instead of presenting disconnected alert strings, Wazuh tags security events with formal Tactics (the attacker's overarching objective) and Techniques (the specific technical mechanism utilized). This enables SOC analysts to reconstruct adversary attack chains across the intrusion lifecycle.

Adversary Tactic	Technique ID	Technique Name	Correlated Wazuh Alert Event
Initial Access	`T1078.003`	Valid Accounts: Local Accounts	Successful interactive login via SSH outside normal shift baseline.
Credential Access	`T1110.001`	Brute Force: Password Guessing	Rapid sequence of failed SSH logins from single IP address (`sshd`).
Persistence	`T1098`	Account Manipulation	FIM detected unexpected modification to `/etc/passwd` and `/etc/shadow`.
Privilege Escalation	`T1548.003`	Abuse Elevation: Sudo / Sudo Caching	Sudo command execution by unprivileged user running `/bin/bash`.
Defense Evasion	`T1070.002`	Indicator Removal: Clear Linux Logs	File truncation event detected on `/var/log/auth.log` by syscheck.

7.1 SOC Dashboard Matrix Visualization

In the Wazuh Dashboard, the *MITRE ATT&CK* tab provides a visual heatmap overlay of the cyber kill chain. During the lab simulation:

- **Credential Access** displayed the highest event count (3,110 hits) driven by automated SSH brute-force testing.
- **Defense Evasion** and **Persistence** elevated from zero to active status following the manual file tampering executed in Section 4.
- Clicking any active cell opens drill-down views displaying the contributing agents, source IP coordinates, and raw event evidence.

8. Automated Active Response & Threat Containment

Passive alerting requires human SOC analysts to review logs before taking defensive action. Wazuh's *Active Response* framework enables closed-loop automated remediation: when an alert crosses a specified severity threshold or matches a defined rule, the Manager signals the target agent to execute local containment scripts (e.g., dropping firewall traffic, revoking a session, or quarantining a file).

8.1 Configuring Active Response Rules (ossec.conf)

To neutralize automated SSH brute-force attacks, an Active Response rule was authored on the Manager, pairing the built-in `firewall-drop` script with rule `5712` (SSH brute-force detection).

LISTING 8.1: ACTIVE RESPONSE CONFIGURATION IN OSSEC.CONF

```
<!-- Declare the executable command --> <command> <name>firewall-drop</name>
<executable>firewall-drop</executable> <timeout_allowed>yes</timeout_allowed> </command> <!--
Bind command to SSH brute force alerts with a 10-minute ban --> <active-response>
<command>firewall-drop</command> <location>local</location> <rules_id>5712</rules_id>
<timeout>600</timeout> </active-response>
```

8.2 Attack Simulation & Dynamic Mitigation Proof

From the attack node (`10.204.133.205`), a high-speed Hydra attack was launched against the Ubuntu agent. Upon hitting 8 failed logins, the Wazuh agent triggered the response script, dynamically updating `iptables`:

LISTING 8.2: AGENT ACTIVE RESPONSE EXECUTION LOG

```
ubuntu@target-srv:~$ sudo tail -f /var/ossec/logs/active-responses.log 2026/09/20 12:45:10 ac
tive-response/bin/firewall-drop: INFO: Starting: -a 10.204.133.205 -t 600 2026/09/20 12:45:10 activ
e-response/bin/firewall-drop: INFO: Added iptables rule: DROP 10.204.133.205 # Inspect iptables sta
te on the protected Linux endpoint ubuntu@target-srv:~$ sudo iptables -L INPUT -v -n | grep 10.204.
133.205 24 1440 DROP all -- * * 10.204.133.205 0.0.0.0/0
```

The Hydra attack process terminated with connection timeouts, proving autonomous threat neutralization within milliseconds of threshold violation.

9. Consolidated Security Findings & Forensic Matrix

The deployment and testing sequence evaluated multiple telemetry sources across the home lab cluster. The primary detections and their operational implications are summarized below:

#	Telemetry Source	Security Event Observed	XDR Analysis & Impact	Severity
1	Syscheck (FIM)	Tampering in `/etc/ssh/sshd_config`	Unauthorized root login enablement detected in real time with whodata attribution.	HIGH
2	Vulnerability Detector	Unpatched OpenSSH (CVE-2023-38408)	CVSS 9.8 Remote Code Execution vulnerability present on standard SSH daemon.	CRITICAL
3	Rootcheck	Anomalous file in `/dev/.hidden_backdoor`	Concealed file matching common rootkit persistence patterns detected during sweep.	HIGH
4	Active Response	Autonomous firewall drop for 10 min	Offending IP `10.204.133.205` blocked via `iptables` after 8 brute-force attempts.	RESOLVED
5	Windows Eventlog	Anomalous user creation via PowerShell	EventID 4720 captured on Domain Controller, mapped to MITRE T1136 (Create Account).	HIGH

9.1 Infrastructure Hardening & Production Recommendations

- **Harden Active Response Scripts:** Ensure response scripts include rate-limiting and administrative IP whitelisting to prevent denial-of-service via spoofed source addresses.
- **Enforce Automated Package Remediation:** Implement automated patching policies for packages identified with Critical CVSS >= 9.0 vulnerabilities.
- **Fine-Tune FIM Whodata Auditing:** Target `whodata` tracking exclusively on sensitive credential and binary stores to minimize Linux kernel audit subsystem overhead.
- **Cluster Redundancy:** Migrate single-node lab deployment to a multi-node Wazuh Manager cluster with load-balanced agent communication (HAProxy).

10. Conclusion

This practical laboratory experiment successfully validated the end-to-end deployment, telemetry gathering, and threat response capabilities of the open-source Wazuh SIEM & XDR platform. By instrumenting heterogeneous Linux and Windows Server endpoints, the lab verified real-time file integrity monitoring, identified unpatched kernel and OpenSSH vulnerabilities, and detected rootkit persistence indicators. Furthermore, correlating raw alerts against the MITRE ATT&CK matrix provided clear behavioral context, while the active response framework autonomously contained brute-force attacks via dynamic packet filtering. The results confirm that combining endpoint-level telemetry with centralized rule correlation provides robust detection and incident response for modern security operations.

11. References

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Laboratory Course Convergence

This experiment marks the culmination of the endpoint and security telemetry series. Starting with surface discovery in Nmap, vulnerability identification in Nessus, traffic pattern inspection in Snort, wire analysis in Wireshark, and centralized log searching in Splunk, this Wazuh deployment establishes endpoint visibility, automated mitigation, and closed-loop XDR defense.